

[5351] - 109

ENGINEERING

Engineering Mechanics

(2015 Pattern) (Semester - II)

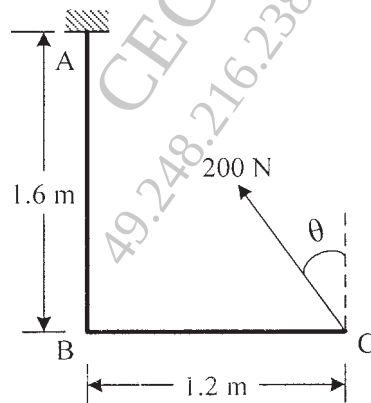
Time : 2 Hours]

[Max. Marks :50

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagram must be drawn wherever necessary.
- 3) Figure to the right indicates full marks.
- 4) "Assume suitable data, if necessary and clearly state."
- 5) Use of nonprogrammable electronic pocket calculator is allowed.

- Q1) a)** The lever ABC fixed at A shown in Figure is subjected to a 200 N force at C at $\theta = 30^\circ$. Find the moment of this force about A. Also find the value of θ for which the moment about A is Zero. [6]

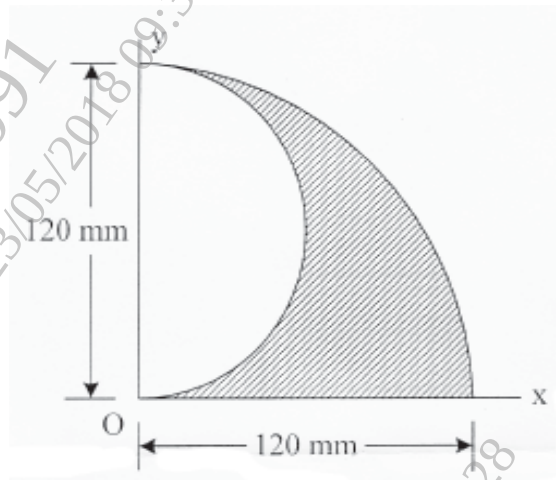


- b)** The motion of particle is defined by, $x = t^3 - 6t^2 + 9t + 5$, where x expressed in meter and t in seconds. Determine the time at which velocity becomes Zero. Also determine velocity and acceleration at $t = 5$ s. [6]

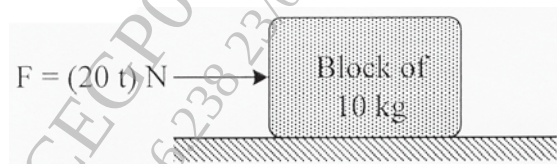
OR

P.T.O.

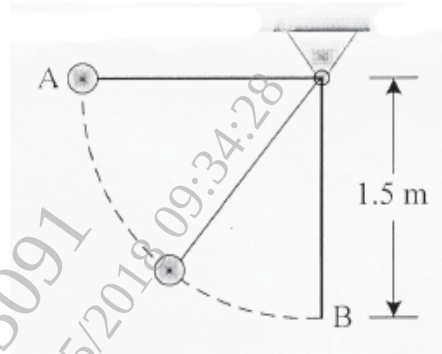
- Q2) a)** Locate the centroid of the plane area as shown in Figure with respect to origin O. **[6]**



- b)** The 10kg block is subjected to the force shown in Figure determine its velocity when $t = 2s$ if $v = 0$ when $t = 0$. **[6]**

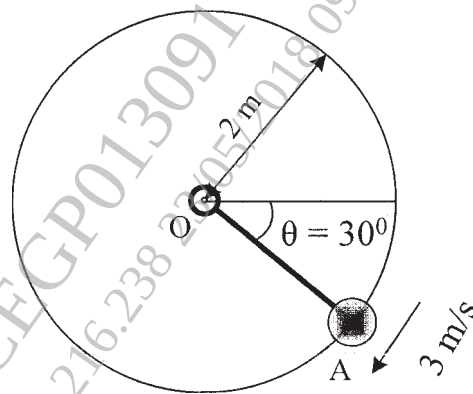


- Q3) a)** A motorist is traveling on a curved section of highway of radius 762m at the speed of 96 kmph. The motorist suddenly applies the brakes, causing the automobile to slow down at a constant rate. Knowing that after 8s the speed has been reduced to 72 kmph, determine the acceleration of the automobile immediately after the brakes have been applied. **[6]**
- b)** The 2kg pendulum bob is released from rest when it is at A as shown in Figure. Determine the speed of the bob when it passes through its lowest position B. **[6]**

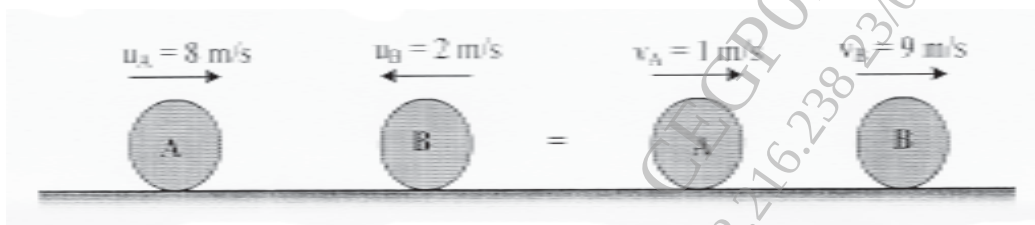


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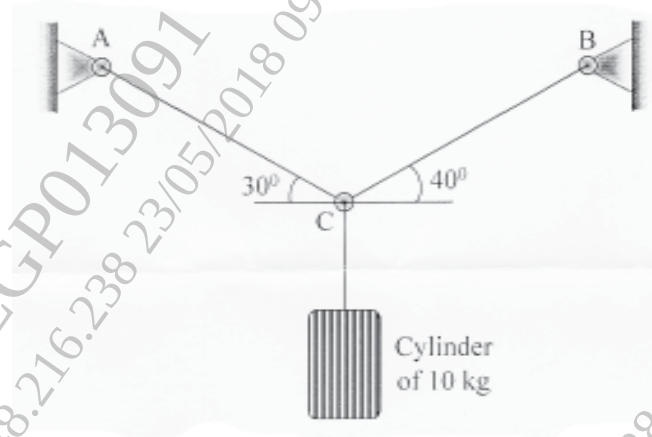
- Q4) a)** If the 10kg ball has a velocity of 3 m/s when it is at the position A as shown in Figure along the vertical path, determine the tension in the cord and the tangential component of acceleration of ball at this position. [6]



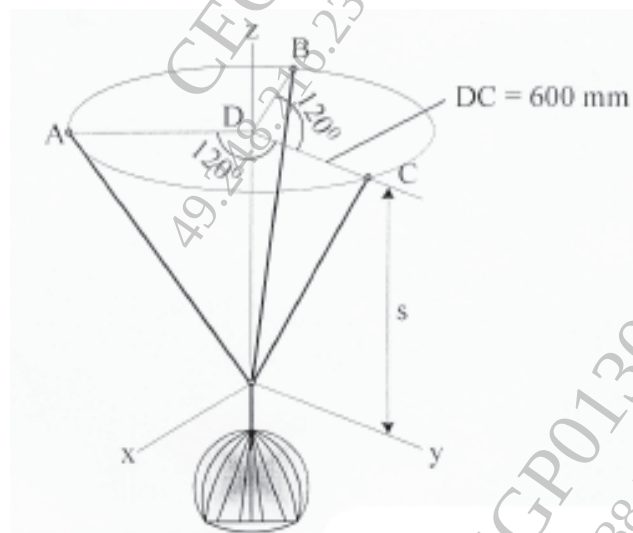
- b)** Determine the coefficient of restitution e between two identical balls A and B. The velocities of balls A and B before and after impact are shown in Figure. [6]



- Q5) a)** Determine the tension developed in wires CA and CB required for equilibrium of the 10kg cylinder as shown in Figure. **[6]**

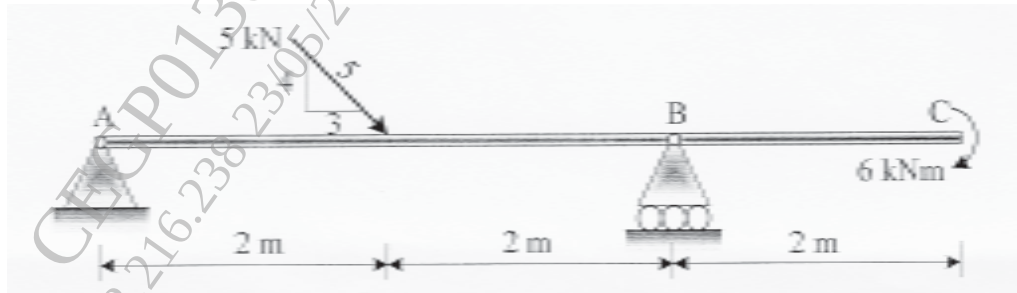


- b)** The 10kg lamp shown in Figure is suspended from three equal length cords. Determine its smallest vertical distance s from the ceiling if the force developed in any cord is not allowed to exceed 50N. **[7]**

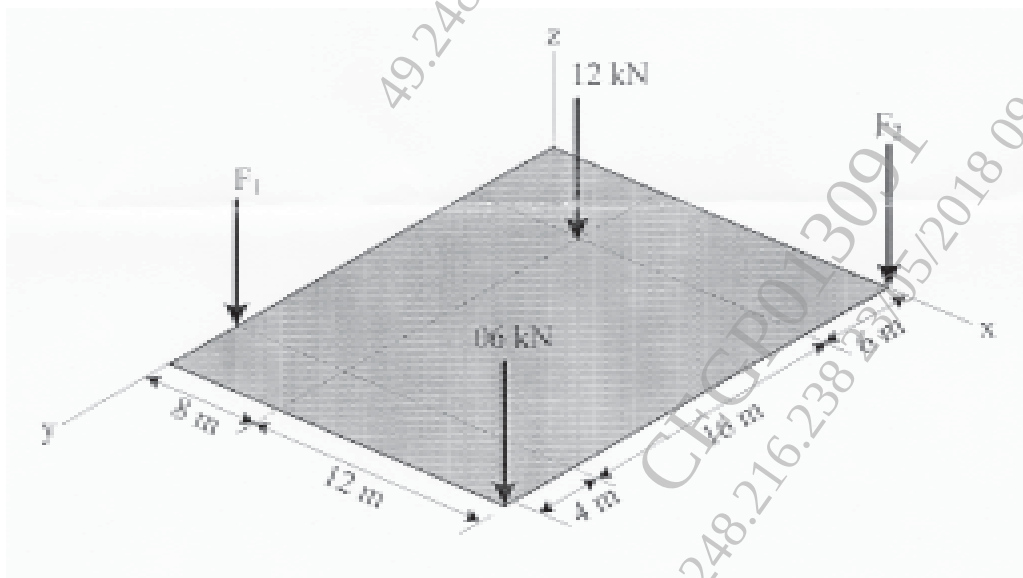


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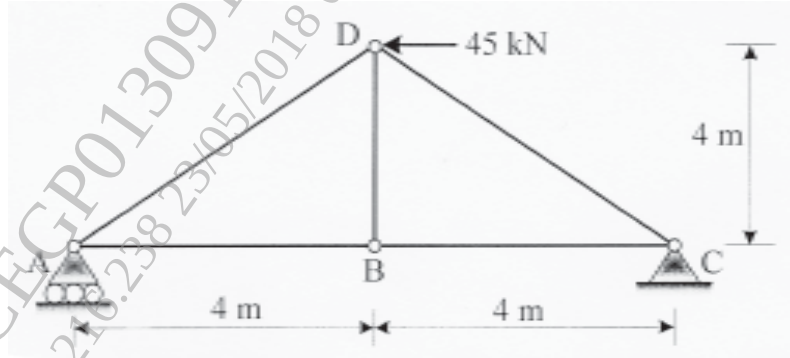
- Q6) a) Determine the horizontal and vertical components of reaction at the supports for the beam as shown in Figure. [7]



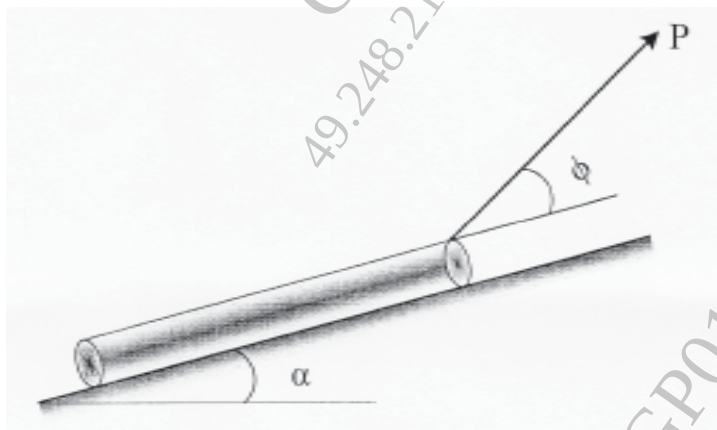
- b) The building slab is subjected to four parallel column loading shown in Figure. Determine F_1 and F_2 if the resultant force acts through point (12m, 10m). [6]



- Q7) a)** Determine the forces in each member of the truss shown in Figure. State if the members are in tension or compression. **[6]**

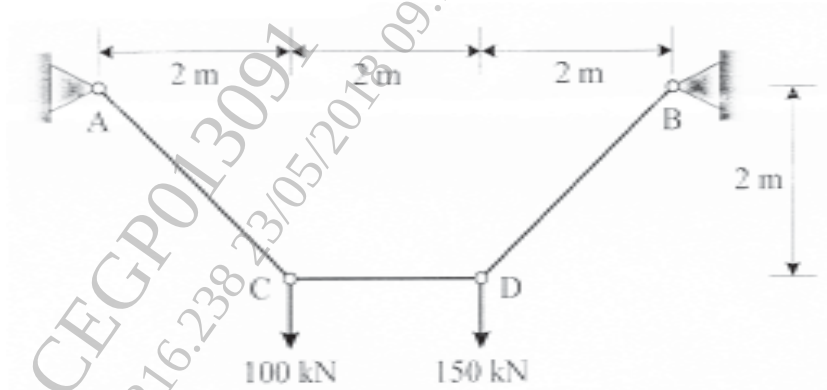


- b)** The pipe of weight W is to be pulled up the inclined plane of slope α using a force P shown in Figure. If P acts at an angle ϕ , show that for limiting condition $P = \sin(\alpha + \phi) / \cos(\phi - \theta)$ where θ is the angle of static friction $\theta = \tan^{-1} \mu_s$. **[7]**



OR

- Q8) a)** For the cable AB as shown in Figure, find the reaction at supports and tension in each segment. [7]



- b) Determine the maximum horizontal force P that can be applied to the 12 kg hoop without causing it to rotate as shown in Figure. The coefficient of static friction between the hoop and the surfaces at A and B is, $\mu_s = 0.2$. Take $r = 300$ mm. [6]

